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4G to 6G: disruptions and drivers for optical access [Invited]

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The exponential growth of diverse smart mobile applications such as smart monitoring, holograms, and autonomous vehicles are driving mobile technologies toward a more intelligent and software-defined communication system. This growing demand has led to the next giant leap in next-generation wireless communication technology, i.e., sixth-generation (6G) mobile technology, which anticipates providing 1 Tbps data rates and ultralow latency over ubiquitous 3D coverage areas. However, the transport network, which connects hundreds of thousands of cell sites and the network core to enable intelligence, virtualization, and other key features of 6G, has not advanced enough to cater the dense cell deployment expected in future 6G networks. Because many aspects of 6G remain undefined, this provides us with the opportunity to take optical transport networks into design consideration to realize the benefits that 6G has to offer. To this end, in this paper, we present a comprehensive view on the role of optical access networks in supporting fourth-generation (4G), fifth-generation (5G), and beyond wireless technologies. In particular, we discuss design and deployment strategies and their challenges when using optical access as a transport solution during the evolution of wireless access technologies. For this purpose, we first identify how an optical access network is used to support high-capacity transport networks in 4G, followed by the challenges that 5G brought into optical transport networks and its diverse solutions. We conclude the paper by providing insights into how an optical transport network can be designed to support 6G.

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1. INTRODUCTION

The usage of smart technologies to make our lives easier and to protect our environment has significantly increased over the past decade. Cisco predicted that the number of connected devices will reach 500 billion by 2030 as driven by the Internet of Things (IoT) paradigm, which will be significantly greater than the expected world population [1]. Providing connectivity for billions of devices that supports diverse applications with their stringent quality of service (QoS) requirements such as high throughput, low latency, and high reliability is continuing to be a major challenge in modern communication networks. To address this ever-growing issue, mobile technology continues to evolve toward better and smarter technologies. While fifth generation (5G) is maturing and currently undergoing active deployment [2–6], academia and industry have recently started working on the sixth-generation (6G) mobile technology [7–11]. The white paper released by the 6G Flagship in September 2019 is the first document to map out the 6G vision for "ubiquitous wireless intelligence by 2030" [12]. Leading telecommunications research and development companies, e.g., Ericsson [13], Nokia Bell Labs [14], and Samsung Research [15], have subsequently released their vision for 6G.

Driven by a variety of new mission-critical applications such as industrial automation and control, vehicular platooning, immersive virtual reality (VR), augmented reality (AR), extended reality (XR), smart grid, and the IoT, future wireless networking infrastructure is required to provide ultrareliable low-latency communication (uRLLC), with end-to-end latency in some cases not exceeding 1 ms and reliability of the order of 99.9999% or even higher [3]. Some uRLLC use cases also simultaneously require high spectral efficiency (e.g., immersive VR) or massive connectivity (e.g., IoT). Certain applications, such as those used in remote surgeries, will require reliable communication that supports submillisecond latency and extremely low error rates, while IoT applications require ubiquitous connectivity and, in some cases, energy efficiency for extended battery life. The stringent constraints imposed by such applications require novel wireless communication system designs in order to attain the optimal trade-off among latency, reliability, throughput, and energy efficiency.

To support these applications and requirements, 6G mobile technology expects to provide data rates of more than 1 Tbps and ultralow latency over ubiquitous 3D coverage areas. User traffic generated by these applications arrives at the wireless cell sites and is carried over to the network core via the transport network. However, while significant focus has been directed toward wireless technologies with increasing user demand, the transport network has not received the same attention despite its key role in end-to-end data transmission. Although the transport network is well equipped in terms of capacity through the deployment of optical fiber, the networking, reconfigurability, deployment cost, and scalability aspects of the transport network have not been evaluated properly to cater for the dense cell deployment expected in 5G and beyond networks. Consequently, in fourth-generation (4G) to 6G wireless access network evolution, the high-capacity reconfigurable transport network, which links the hundreds and thousands of cell sites with the network core, has been identified as a potential bottleneck in high-performance, cost-efficiency, and network deployment.

With the increasing demand for data-hungry applications delivered through wireless networks, optical access technologies are becoming a cornerstone of next-generation wireless broadband access to meet their stringent demands for high bandwidth and ultralow latency. As such, moving forward from 5G, optical transport networks that carry all the wireless data from wireless cells toward the network core also need to be carefully designed and implemented simultaneously with the evolution of wireless technologies in order to deliver all the ingenuity features that next-generation wireless networks have to offer.

To reduce deployment and operational costs, enhance the QoS and quality of experience (QoE), allocate resources efficiently to improve network throughput, and reduce energy consumption, we need to coalesce the benefits of wireless and optical access technologies into a single platform that functions as a single entity. Many mechanisms, technologies, and frameworks have been proposed and implemented to support this unified operation and deployment of integrated optical and wireless access networks. Therefore, in this paper, we discuss how optical transport networks have evolved and integrated with wireless networks to support the wireless access network starting from 4G to 5G and beyond with specific focus on the deployment perspective of the network. In this paper, we also discuss each major planning and dimension challenge we face in 4G and 5G and beyond and the approaches that have been proposed to overcome those challenges. For example, the deployment of economical small cells has been used since 4G to increase the per-user bandwidth, per-area bandwidth, and network coverage. Even though these small cells bring many benefits, they also lead to a new set of challenges due to the dense deployment of small cell, especially related to the deployment of optical transport networks.

The rest of the paper is organized as follows. Section 2 discusses the evolving wireless access technology landscape, including supporting applications, technological advancements, and key performance indicators. We then discuss the role that optical networks play in supporting each of these wireless access networks or technologies. In particular, the drivers behind choosing optical access networks to support 4G are discussed in Section 3. Section 4 then discusses the challenges facing 5G and how an optical access network can be used as a transport solution to overcome those challenges. Section 5 briefly discusses the role that optical networks will play in supporting 6G and converged network architecture that is envisioned to support 6G before we conclude our discussion in Section 6.

2. EVOLUTION OF WIRELESS ACCESS TECHNOLOGIES

To demonstrate the important role optical transport technologies play in supporting current and future wireless access technologies, in this section, we first provide a brief overview of the evolution of wireless technologies, from legacy first-generation (1G) in the 1980s to the current 5G and upcoming 6G standard over the next decade. For more detailed information, we refer the reader to [16–21]. Figure 1 summarizes this evaluation of wireless access technologies with emphasis on key enabling technologies, modulation schemes, achievable data rates, and applications throughout this evolution.

The 1G mobile communication system was based on the Advance Mobile Phone System (AMPS) introduced by Bell Labs in 1981. 1G used systems such as the Nordic Mobile Telephone (NMT), Total Access Communication Systems (TACS), and European TACS (E-TACS) to facilitate voice transmission [18,21]. These systems only supported voice calls using analog signals with bit rates in the range of 4.8–9.6 Kbps and frequencies ranging between 800 and 900 MHz with a 10 MHz bandwidth. Frequency modulation (FM) was used with frequency division multiple access (FDMA). 1G is prone to noise, attenuation, interference, weak security, poor voice quality, large-sized phones, poor hand-off reliability, frequent call drops, poor battery life, small distance coverage, and limited number of systems. Amidst these shortfalls, however, 1G holds its importance as the first step toward this continuing evaluation we witness today.

The second-generation (2G) telecommunication technology, introduced in 1992, represents a shift from analog to digital communication. Digital signals can cover a larger distance and area and are less prone to noise and interference. 2G is capable of supporting data rates in the range of 14.4–64 Kbps and enables services such as Short Message Service (SMS) and Multimedia Messaging Service (MMS) [18]. 2G uses code division multiple access (CDMA) and has been globally accepted as the Global System for Mobile Communications (GSM) standard, which quickly became the base standard for further development in wireless communication technologies. For the first time in 2G, Internet access became possible (albeit at a much lower rate), and data are digitally encrypted. In addition, services such as roaming are also possible in 2G. However, 2G still suffers from limitations such as low data rate, limited mobility, limited users, and hardware capability. To address some of these limitations, 2.5G and 2.75G were later introduced in 1997. 2.5G introduced the general package radio service that used packet-switched

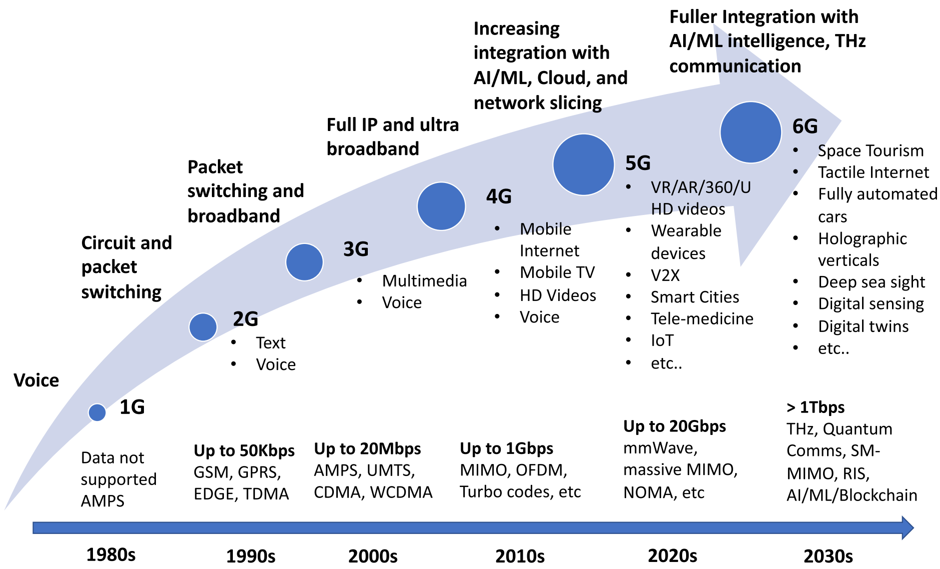

Fig. 1. Brief overview of the evolution of wireless access technologies from 1G to 6G.

technology (capable of 56–164 Kbps data rate) in addition to the circuit-switched GSM network. 2.75G subsequently introduced Enhanced Data rates for GSM Evolution (EDGE) for a much improved data rate, supporting a data rate up to 473 Kbps. EDGE uses an 8 phase-shift keying (8PSK) modulation scheme for CDMA to deliver a higher data rate. CMA2000 was later introduced to support a higher data rate for CDMA networks.

The third-generation (3G) mobile technology introduced the Universal Mobile Terrestrial/Telecommunication System (UMTS) in 2001 [18]. It supports a data rate of 384 Kbps for mobile users, and video calls/multimedia chats were supported for the first time on mobile devices. In 3G, a greater number of users can be simultaneously supported by a radio-frequency bandwidth, with high-speed mobile access to services using the Internet Protocol (IP). 3G uses FDMA and time division multiple access to transfer voice data and nonvoice data with less complexity, larger capacity, and broadband access, all with a higher data rate and at a lower cost. As a result, applications such as video calls, music downloads, gaming, web browsing, emails, and instant messaging are supported over the same network simultaneously. In 2001, 3.5G was introduced with a High Speed Downlink Packet Access (HSDPA) and High Speed Uplink Packet Access (HSUPA). These systems were capable of data rates up to 2 Mbps. In 2004, 3.75G was introduced with High Speed Packet Access Plus (HSPA+), which later became 3.9G with Long-Term Evolution (LTE). 3.9G provides higher QoS, higher data rate, and higher coverage area at a lower cost compared with its successors. A substantial number of users are also supported, along with a more diverse range of applications, such as on-demand video, peer-to-peer file sharing, and composite Web services. In 3.9G, multiple input multiple output (MIMO) with 64 quadrature amplitude modulation (QAM) was implemented, which enables 3.9G to support data rates up to 30 Mbps downlink and 11 Mbps in uplink.

In the 4G mobile communication system, much higher data rates (up to 1 Gbps for stationary users and 100 Mbps for mobile users) are achieved with enhancements to the LTE technology, LTE advanced. The significant increase in bandwidth and reduction in latency supported by 4G, along with advances in smart phone development has resulted in a surge of bandwidth-hungry applications such as high-definition multimedia streaming, real-time video calls, online gaming, etc. [22]. One of the major characteristics of 4G is its shift to full IP-based communication, where all services including voice data are transmitted as IP packets. The all-IP packet core network enables real-time delivery of IP services and enables efficient delivery of packet-oriented multimedia services via the IP Multimedia Subsystem (IMS). Complex modulation schemes such as MIMO, orthogonal frequency division multiplexing, and turbo codes, along with carrier aggregation techniques are used in 4G to multiply the uplink and downlink bandwidth capacities. One of the key features that the 4G system introduces is the reduction of cell sizes [23]. The coexistence of small cells in heterogeneous networks consisting of macro cells became a key feature in achieving the higher data rates and network throughput targets of 4G.

Today, 5G mobile communication technology is being actively deployed in many countries. 5G promises higher-bandwidth capacities, lower latency, and higher reliability for emerging time-sensitive and mission-critical applications. Notably, 5G promises to enable 10-100X data rate improvement, 1000X bandwidth per unit area improvements, 100% coverage, and 10-100X latency enhancements compared with its predecessor 4G [4,18]. Earlier deployments of 5G networks will be available in non-standalone and standalone modes. In non-standalone mode, both LTE spectrum and 5G-New Radio (5G-NR) spectrum will be used simultaneously. Control signals will be transmitted via the LTE core network. In standalone mode, a dedicated 5G core network with higher bandwidth using the 5G-NR spectrum will be used. A sub-6-GHz spectrum is used in the initial deployments of 5G networks. One of the significant innovations in 5G is the use of millimeter waves (mmWave) and unlicensed spectrum for data transmission and complex modulation techniques to achieve significantly higher data rates. Key enabling technologies of 5G include massive scale of cell densification using smaller cellular coverage areas, massive MIMO, edge computing, beamforming, convergence of WiFi and cellular, software-defined networking (SDN)/network function virtualization, network slicing, channel coding (with polar codes), and reconfigurable intelligent surfaces (RISs). The well-known "5G triangle" captures the three key innovations of 5G, namely, enhanced mobile broadband (eMBB), uRLLC, and massive machine-type communication (MTC). These innovations, in conjunction with technologies such as AI and IoT, will significantly expand the landscape of mobile communication applications. As applications and industry verticals are no longer limited to smart phones, 5G opens up vast opportunities in VR/AR/XR, 360 videos, wearable devices, vehicle-to-everything (V2X) communication, tele-medicine, IoT applications in various industries, and so forth.

The technological applications, use cases, and standards and enabling (and the much anticipated breakthrough) technologies are currently undergoing active research and experimentation to shape the upcoming 6G communication technology. 6G will see a much more "intelligent" communication landscape at the THz spectrum with data rates exceeding 1 Tbps and submillisecond latency [10,24]. A much more optimized and communication-computing converged paradigm is also expected in 6G, where breakthrough technologies such as quantum computing, communication, and networking are expected to play a role in 6G. Currently, there are grand visions for 6G, such as the space-air-ground integrated network [25–27], space tourism, tactile Internet [28–30], autonomous vehicles, holographic communication, deep sea sight, digital sensing, digital twins, and so forth. How these applications will materialize remains a challenge and will be a key driving force for research and development over the next decade.

In conclusion, this discussion demonstrates the evolution of wireless networks fueled by increasing numbers of users and applications with varying QoS requirements. However, wireless technology itself is not sufficient to support all the requirements for end users and applications and relies on key network segments such as transport networks to connect to the rest of the network. As such, the transport network itself has to adapt to cater for high bandwidths and low latency, which has been made possible by optical access networks, as discussed in recent literature. In the following three sections, we discuss the role that optical networks play in supporting 4G, 5G, and 6G wireless access technologies focusing on the design and deployment perspective of the transport network.

3. OPTICAL TRANSPORT AS A CONVENIENT OPTION IN 4G

As discussed in the previous section, the transition from 3G to 4G was motivated by the necessity to make multimedia applications such as videos, voice calls, and Internet access available to users from anywhere at anytime in a secure manner [31]. In supporting such applications, the underlying communication networks had to be improved with high-speed ubiquitous access with data rates in the range of 100 Mbps (mobile users) to 1 Gbps (stationary users) and latency less than 50 ms [32]. Due to their inherent high capacity, data rate, and low losses, optical access networks were identified as viable candidates for the transport segment of the network. However, when using optical access as a transport network solution, network operators were faced with major challenges such as providing cost-effective 4G small-cell backhaul and providing a fully converged optical-wireless network to support the required QoS of end users.

A. Small-Cell Backhauling

The radio access network (RAN) resides between the end user equipment (UE) and the core network to enable last-mile wireless access to end users. In a typical 4G RAN, multiple macro base stations (BSs) are placed across the access network, with UEs connected to these BSs to access respective services and applications. The resources such as bandwidth available at a macro cell are limited and are shared among multiple UEs served by the macro cell. Although this shared bandwidth was sufficient for voice coverage among users, network operators were faced with new challenges as cellular networks moved toward all-IP data services that were previously delivered through wired communication networks. Not only were the wireless networks required to deliver data-hungry applications, users also expected the same level of quality they would get in wired networks from wireless networks. In order to satisfy these requirements, a simple approach of reducing cell sizes has been identified as an effective solution [23].

The general idea of small-cell development was to deploy small cells close to the UEs in a RAN [33]. The small cells are typically positioned as an underlay to existing macro cells in areas where there is not sufficient coverage to deliver the aforementioned data-hungry applications. As discussed in [34], as the small cells have a much shorter transmitter–receiver distance, it results in lower transmit power and higher signal-to-interference-plus-noise ratio, which leads to improved reception and higher capacity. On the other hand, with densely deployed small-cell networks, there was an inevitable increase in volume of data generated at RANs. Tasked with transporting this high volume of data to the core network while satisfying the QoS requirements, network operators identified optical networks as a backhaul solution for small-cell networks due to their high-capacity and low-loss characteristics.

One of the key challenges with the increasing small-cell deployment was the increased capital expenditure and time associated with deploying optical backhaul real estate [35,36]. However, telecommunication operators have already deployed optical access networks such as fiber to the node (FTTN) and fiber to the home as means of providing high-speed broadband access to customers. The spare fiber in such networks can be used as optical backhaul for small-cell networks to reduce deployment cost. However, as reported in [37], the dark fiber in such networks is a scarce resource that would not be used efficiently if used as a backhaul link for small-cell networks. Alternatively, the cost associated with fiber backhaul can be further reduced by deploying passive optical networks (PONs) such as Ethernet passive optical networks and gigabit passive optical networks (GPONs) on top of the existing fiber infrastructure, as one fiber can be shared among multiple small-cells in PONs. As such, significant research has been carried out to investigate how to effectively and optimally leverage the resources associated with existing infrastructure to provide optical backhaul for small-cell networks.

For example, in [38], the authors presented an optimization framework that can be used to plan a cost-minimized GPON backhaul deployment for small-cell networks. When the proposed framework is used to plan a small-cell backhaul network for a data set derived from a real network, it was shown that the deployment cost can be halved compared with existing FTTN used as a backhaul link. It is important to note that optimization criteria must be strategically planned, as considering just the cost of backhaul deployment can hinder more important aspects such as coverage. To address this issue in a concise manner, the studies reported in [37,39] have also incorporated additional constraints such as capacity, coverage, and energy in the optimization frameworks.

B. Optical-Wireless Convergence

The optical-wireless convergence aims to combine cost-effective high-data-rate capabilities of optical access networks with the mobility of wireless networks. When an optical access network is used as a backhaul network, to guarantee the QoS requirements of end users, special attention should be given to the converged network architecture and the resource-handling mechanism implemented, as optical and wireless networks have different requirements and use different QoS mechanisms. The left side of Fig. 2 illustrates a typical optical-wireless convergence network used in 4G networks. The architecture shown here depicts a decentralized RAN architecture, where the processing of wireless signal is carried out at individual base stations. As shown in the figure, point-to-point optical fiber links or point-to-multipoint PON can be used to connect the base station to the aggregate node directly. However, when using a PON topology to connect the base station with the aggregate node, the optical line terminal (OLT) is placed at the aggregated node and the optical network unit (ONU) at the base station. The optical fiber link from the OLT goes through a 1:n optical splitter to reach n number of base stations. The split ratio n can be determined based on the data rate requirements of the base station and the line rate of the PON technology in use.

Although implementing wireless and optical technologies in the transport and access networks seems straightforward, as shown in Fig. 2, careful consideration should be given when integrating two networks that have their own technologies and protocols, to ensure seamless connectivity between different network entities in 4G. One of the challenges in providing optical backhaul for 4G using PON was to satisfy the architectural requirements of 4G. For example, the LTE network requires implementing the intercommunication between base stations to facilitate the proper operation of wireless techniques such as coordinated multipoint. In LTE, this is achieved through the X2 interface, which is the direct interface between neighboring base stations. However, when using PON, this direct communication between neighboring base stations is implemented via the OLT. To overcome this challenge,

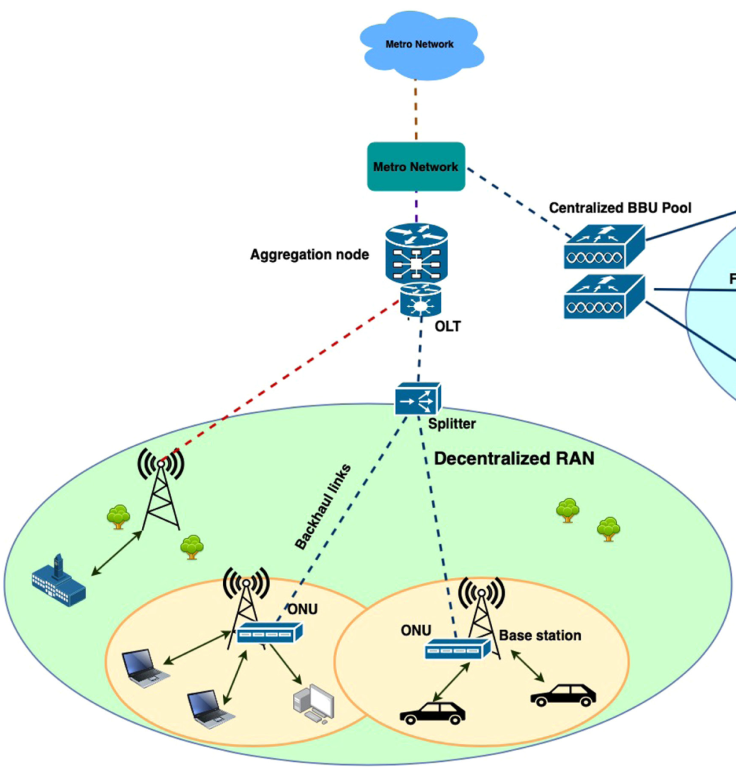

Fig. 2. Optical-wireless converged network and RAN architectures.

researchers have proposed different PON-based architectures to provide such a communication either via the splitter or via base stations that are connected with ring-based optical wavelength division multiplexing (WDM) PON [40–42].

In addition to satisfying the connectivity requirements in the 4G networks, the converged networks should also satisfy the QoS requirements of the intended applications delivered over 4G. For this purpose, resource, more specifically bandwidth, allocation in both segments of the converged network should be carefully managed and evaluated. Such efforts include taking QoS requirements of different applications such as latency, throughput, and reliability into consideration and designing bandwidth allocation algorithms to satisfy these requirements. To date, diverse bandwidth allocation algorithms have been proposed to guarantee the required QoS of the end users [43,44]. For example, coordinated resource allocation algorithms have been proposed to allocate resources in both domains simultaneously, leveraging the benefits of the knowledge shared within the networks. Some previous works have looked at how the OLT can be integrated into a central controller that can allocate resources globally for optical and wireless networks to satisfy the network's QoS requirements [43].

4. OPTICAL TRANSPORT FOR SOLVING MAJOR CHALLENGES IN 5G

In Section 3, we discussed the role of the optical network as a transport network for 4G and its associated challenges. In this section, we discuss the role an optical network plays in solving major challenges in 5G, which is considered to be the biggest evolution in wireless access technologies thus far. In Section 2, we explained that 5G consists of several key attributes to achieve the expected key performance indicators such as 1 ms latency and 10 Gbps data rate. One of the important features is that 5G is equipped to reach these targets on a massive scale of cell densification. Because we have a limited wireless spectrum, one technique to enable high data rates is to reduce the cell radius and have hundreds and thousands of smaller cells in the network. Apart from having small cells in the network, 5G also has other key features such as massive MIMO and beamforming to increase capacity supported by a single cell [4].

5G networks consist of a hundred times more high-capacity cells in the network compared with 4G, which presents the challenge of connecting these cells to the rest of the network. For this purpose, we need to deploy a high-capacity transport link for each of these small cells. In 4G, we used the term "backhaul" more often for the connection between the base station and the core network, as we have standalone base stations at the cell site (the signal processing is carried out at the base station and sends the packetized data to the core network via the backhaul link). However, we have different network architectures in 5G RAN. As shown in Fig. 2, 5G RAN consists of different types of RAN architectures. One of the main architectures proposed for 5G is the cloud/centralized RAN (C-RAN), where the wireless signal processing is centralized in a base band unit (BBU), which is placed at a central location [45]. In this architecture, only the antennas are placed at the cell site, which is called the "radio remote head" (RRH) or "radio unit" (RU), and a pool of BBUs is typically colocated at a central office (CO). The transport link that connects the BBU and RRH/RU is commonly called the "fronthaul link." In addition to C-RAN, other RAN architectures have also been proposed for 5G, including decentralized RAN (D-RAN), which is similar to the architecture of RAN used in 4G and fog-RAN (F-RAN). In F-RAN, the RUs are also equipped with caching enabled devices to facilitate fog computing at the edge.

In native C-RAN, a fronthaul network operates on a Common Public Radio Interface (CPRI) over optical links. A CPRI is a digital radio over fiber technology and sends IQ data of the baseband signals. CPRI requires a point-to-point fiber connectivity, as it requires a larger bandwidth. The alternative to CPRI, analog radio over fiber (ARoF) technology, was proposed in [46]. The optimal placement of elements in C-RAN to achieve cost-effectiveness is challenging, as the network consists of hundreds of small cells. Reforming this challenge into the problem in placing BBU in the network was investigated in [47]. In this work, an optimization framework was formulated to optimally place the BBUs in COs, assign an optimal number of wavelengths, and allocate wavelength routes to minimize the number of BBU pools and fiber used. Another network dimensioning framework is proposed to plan the 5G native C-RAN where the main objective was to reduce the total deployment cost of 5G and optical fronthaul [48]. In this work, physical layer split, ARoF, and CPRI are considered potential fronthaul technologies for the deployment.

In a typical 5G C-RAN architecture, radio frequency (RF) signals are digitized and sent to the centralized BBU using CPRI. The bandwidth required for sending these digitized signals between the RRH and BBU scales with the physical parameters of an antenna, including number of antenna ports and wireless bandwidth in use [49]. 5G utilizes a large number of antenna ports to support beam forming and massive MIMO. As a result, to send the digitized signal to the BBU, an RRH requires a high bandwidth fronthaul link. For example, the fronthaul bandwidth required for a 5G system that operates with 64 QAM, a MIMO antenna array of 8×12, 200 MHz of bandwidth, and 96 antenna ports to support average data rates of 1.5 Gbps at a RRH is 885 Gbps [50].

This highlights the need for a high-capacity fronthaul when using native C-RAN for a 5G network. In addition to the capacity, there is a challenge in providing the required latency between the BBU and RRH. This becomes important, as it limits the distance between the BBU and RRH. As a result, some of the functions are placed back in the RRH to reduce the fronthaul bandwidth requirement and delay constraints. There are many different possibilities of functions that can be split between the BBU and RRH. These options are now standardized by the next-generation fronthaul interface [45] into 10 different functional split options. These functional split options and the bandwidth required for each are shown in Fig. 3. Because each functional split option provides different requirements for the connection between the RRH and BBU, the connection is called x-haul rather than fronthaul. As can be seen in Fig. 3, a functional split option determines the bandwidth and also latency requirements of the x-haul. For example, when Option 1 is in use, only 1 Gbps of bandwidth is required in the x-haul, as all functionalities are kept at the cell site, and only packetized processed data are sent to the central controller/BBU. On the other hand, when Option 8 is in use,

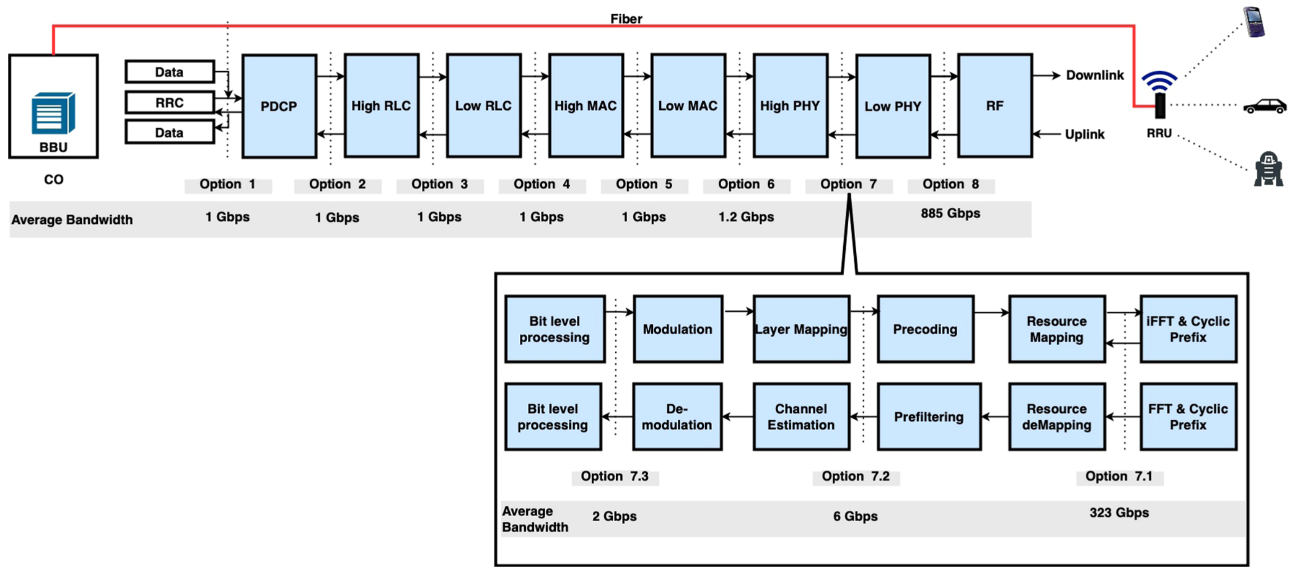

Fig. 3. Next-generation fronthaul interfaces: 5G functional split options.

more than 800 Gbps of bandwidth is required in the fronthaul as all the functionalities are placed at the BBU.

From Option 1 to Option 6, the bandwidth requirement scales with the number of active users and their traffic, as upper-layer functionalities are centralized, and only the physical layer functions are placed in the cell site. However, from Option 7 to Option 8, bandwidth requirement increases exponentially, as the bandwidth depends on the antenna's parameters. Therefore, depending on the functional split option used, different optical technologies and architectures can be used in x-haul. For example, while PON can be used from Option 1 to Option 7.3, point-to-point high-capacity optical technologies are required for Options 7.1 and 8. However, when resource-allocation-based high bandwidth PON such as time wavelength division multiplexing PON is used, careful consideration should be given to solving the timing and latency issues in the transport network through the cooperative transport interface and relevant protocols, which can be used to pass the control messages and bandwidth allocations [51].

These functional split options bring a new set of challenges into planning and dimensioning of 5G and its optical x-haul network. Not only do we need to consider the constraints arising from 5G and its optical transport network, but we also have to consider additional constraints introduced by the functional split options when finding a cost-effective and an efficient deployment solution. Optimal network planning methodologies are now required to recognize a cost-effective solution regardless of the functional split options and optical network technologies used in 5G RAN architecture.

To overcome this challenge, a comprehensive optimization framework is proposed to plan 5G RAN along with its optical x-haul [50]. This framework can be used to jointly plan 5G RAN with its optical x-haul (that uses different functional split options) while satisfying various network requirements, including network coverage and user capacity. The framework can be applied to situations where the network consists of existing fiber infrastructure and hence leverage the resources associated with the fiber infrastructure and also in the situations where existing fiber resources are sparsely located. The framework is developed based on integer linear programming with the objective of minimizing the total deployment cost. Therefore, the objective function consists of the installation and equipment costs of wireless and optical x-haul deployment. The overview of the framework is shown in Fig. 4, which shows the objective function of the framework consisting of different cost components. The framework also comprises several constraints to satisfy the network requirements, including coverage, capacity, and connectivity.

The framework produces optimal fiber routes either selected from existing fiber routes or from given road networks to install new fiber routes and the optimal locations of BBU placement from a given set of CO locations, splitter/WDM MUX placement, and RRH/ONU placement.

The optimal cost and deployment for different deployment scenarios was analyzed by applying the optimization framework to a real data set. For example, Fig. 5 shows the analysis on how the deployment cost varies when the user capacity requirement varies and x-haul consists of different optical network technologies. A cell radius of 200 m and optical technologies such as 10G point-to-point fiber, 10G WDM PON with 1:8 split, and 10G WDM PON with 1:4 split are considered for this analysis. The bar chart in Fig. 5 also shows the cost contributions of the deployment of wireless components, fiber deployment, and fiber-related equipment. As expected, the total deployment cost increases when the capacity requirement of users increases. The cost of fiber installation/usage is the main cost contributor under all the deployment scenarios considered in this analysis. As can be seen, the deployment cost is doubled when the capacity requirement is increased from 50 to 100 Mbps. This is mainly due the fact that we need more cells and more x-haul links when the network has higher-capacity requirements. Those additional x-haul links might require the deployment of new fiber links/paths (involves trenching) and incur higher deployment costs. Further, in Fig. 5, a significant cost difference can be seen when different optical technologies

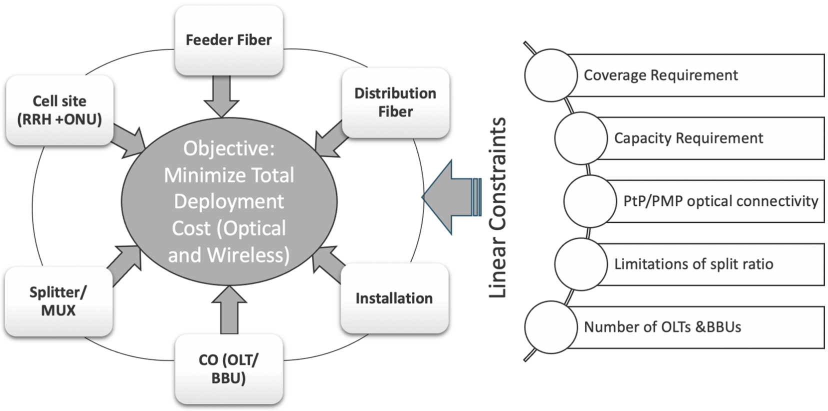

Fig. 4. Generalized optimization framework of 5G RAN planning.

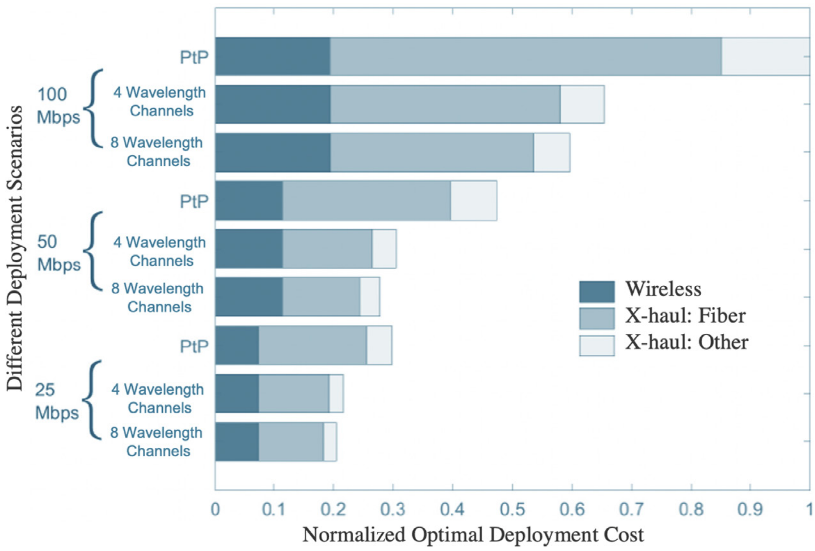

Fig. 5. Optimal cost analysis under different deployment scenarios [50].

are used, especially for high user capacity scenarios. For example, WDM PON-based x-haul can save more than 40% of the total deployment cost compared with 10G point-to-point fiber deployment under 100 Mbps user capacity requirements.

A broad analysis of diverse deployment scenarios showed that 30%–40% of the total deployment cost can be saved when the 5G RAN consists of functional split options from 1 to 6 compared with functional split Option 7.2. Functional split Options 8 and 7.1 resulted in incomparable higher deployment cost, as these options require high bandwidth point-to-point optical fiber links. These generalized optimization frameworks can be used to identify the most suitable deployment strategies that can be deployed in 5G and beyond network deployment, where different functional split options are in consideration.

5. OPTICAL TRANSPORT FOR BREAKING THE BARRIERS IN 6G

In Sections 3 and 4, we discussed the increasingly important role optical networks play in supporting 4G and 5G networks. In this section, we discuss the role that optical networks will play in breaking the barriers in achieving the performance requirements of 6G. As briefly introduced in Section 2, the applications supported by 6G are expected to be more diverse and more far-reaching than ever, with unprecedented demand and requirements for high bandwidth capacities, ultralow latency, ultrareliability, and resiliency. Intelligent communication powered by AI including machine-learning technologies are expected to play key roles in realizing verticals such as digital twins, holographic communications, industrial automation, remote surgery, space exploration, communication, and much more.

Researchers have broadly defined three new types of services that will be supported by 6G: computation oriented communications, contextually agile enhanced mobile broadband communications, and event-defined uRLLC [52]. Although these services are currently loosely defined, it is envisioned that they will require much more stringent requirements, with higher end-to-end service guarantees, and multidimensional key performance indicators such as situational awareness, learning ability, storage cost, computation complexity, and so forth. 6G will therefore need to optimize multiple performance metrics in an increasingly complex and extremely heterogeneous network environment.

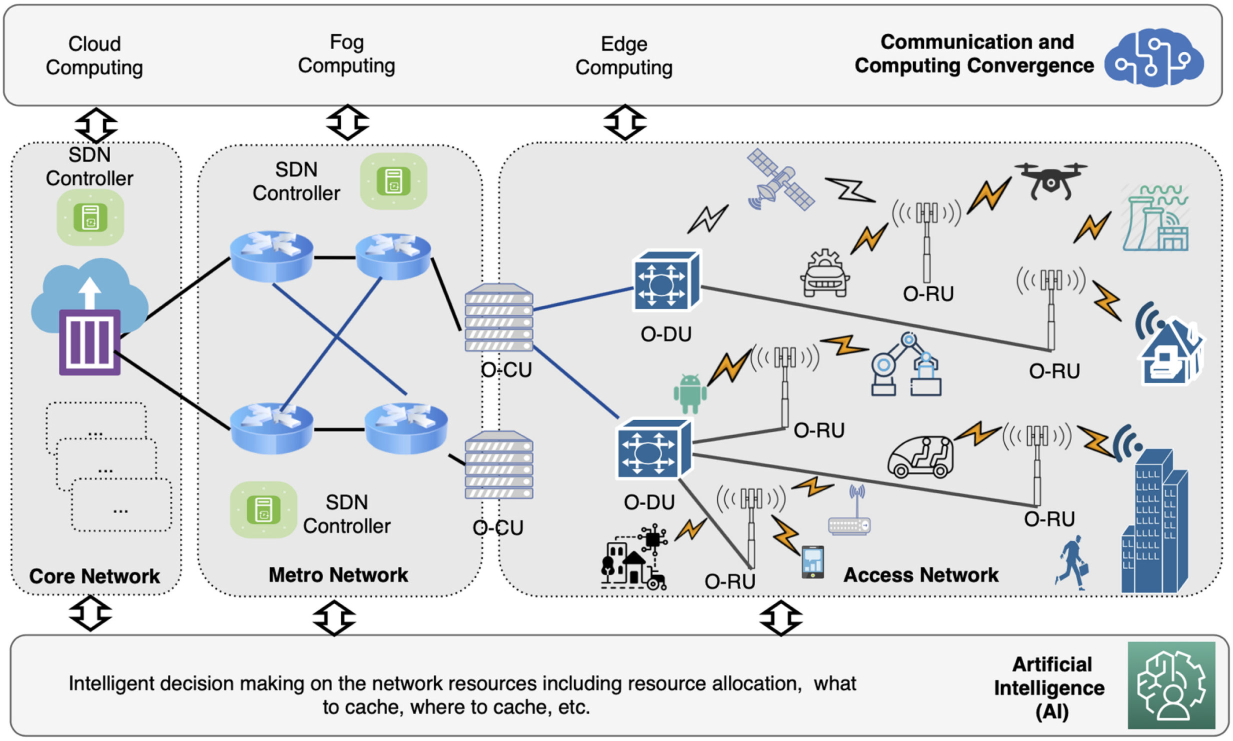

Fig. 6. 6G and optical transport network: envisioned architecture.

To realize these advancements, it is envisioned that 6G's access network will move toward more open and intelligent RAN options. Figure 6 presents the envisioned network architecture of 6G. As shown in Fig. 6, the access network consists of multiple domain networks, including terrestrial and aerial networks. In particular, to provide 3D coverage and connect billions of devices, 6G will employ multiple network types, moving beyond the standard terrestrial networks to aerial and high-capacity indoor wireless networks. The access network will consist of architecture similar to open and intelligent RAN (O-RAN) [53]. O-RAN provides interoperability and diverse interconnection among RAN elements, different RAN software, and resource-handling mechanisms designed by different vendors. As shown in Fig. 6, RAN consists of an open RU (O-RU), open central unit (O-CU), and open distribution unit (O-DU), which are placed between the O-CU and O-RU. In this architecture, we will have different functional split options in different segments of the access network. For example, we could have Option 8 between the O-RU and the O-DU and Option 6 between the O-DU and the O-CU [53].

Communication, computation, and caching convergence will also play pivotal roles in realizing the performance targets of 6G. Instead of having each function as a separate entity, 6G will require a holistic approach in managing communication, computation, and caching functions as a single entity with the use of network intelligence. This approach is shown in Fig. 6 as an all-encompassing layer driven by cloud, fog, and edge computing. These computing paradigms will need to be incorporated alongside the AI capabilities in the network's design phase. There will be many processes and many high-speed links in the network, not only to provide connectivity but also to process data and make informed decisions (on how to allocate resources, what and where to cache, etc.) in real time. Therefore, optical networks will be an integral part of 6G rather than an external element. Machine and network intelligence will play key roles in combining all these technologies together to support the requirements of 6G. However, this will introduce many challenges in the optical transport network, where it should be able to guarantee the connectivity of different elements at a diverse level of granularity. More flexible transport networks should be able to create connections dynamically on the fly and provide required resources on demand—not only to support end user requirements but also to provide required connections for diverse services embedded into 6G. Therefore, design, scalable planning, and resource allocation in optical transport networks need to be rethought to support 6G.

6G networks will see a user-centric network architecture largely driven by AI and ML algorithms, where end-points can make autonomous network decisions based on previous network operations [54]. This mechanism will drastically reduce the overhead of communicating to and from the centralized controllers. Optimized network management will be achieved in 6G, with network planning, resource allocation, and decision-making all (in whole or in part) delivered by using insights derived from ML models. ML models can be run in a distributed fashion and process ML algorithms in real time (with submillisecond latency), hence achieving the QoS requirements of many 6G applications.

As discussed in Section 4, network management, maintenance, and diagnostics are cumbersome jobs and often difficult to optimize based on multifaceted requirements. These will be even more complex in 6G, as many constraints and requirements are already in place. With the assistance of ML-driven networks, we can expect to see self-organizing strategies in complex networks, such as self-learning, self-configuration, and self-healing mechanisms [11]. Optimal network resource allocation can also be achieved in wireless and optical domains with self-optimizing networks that adjust and adapt based on network demand and link availability. It is also expected that 6G will leverage the current enabling technologies in SDN to make networks more open, programmable, and self-driving [55–57].

AI and ML technologies are often seen as a silver bullet for solving modern day challenges. However, there remain many challenges in deploying AI and ML strategies in networking. The 10 most crucial challenges of designing and deploying ML technologies for 6G are discussed in [52]. These challenges include optimized protocol interactions from the physical to application layers, dynamic online learning with proactive exploration, distributed, or centralized network control systems, network scalability for global intelligence, efficiency of learning in architecture design and optimization, and standardization of AI-embedded communications. Enhanced security and privacy protection mechanisms will also form an important aspect of 6G design and development considerations.

As with each generation of mobile technology, there is always a growing need for new and significantly enhanced communication protocols to meet new challenges. Owing to the trends, requirements, services, and their associated challenges in 6G, as discussed in Section 2, 6G will require radical new protocol designs. For instance, 6G must introduce new, AI-driven protocols for wireless and optical segments of the network to carry out signaling, scheduling, and coordination that can replace conventional 5G protocols, which typically rely on predetermined network parameters and rigid frame structures. These new 6G protocols will, in contrast, continuously adapt to the current and projected state of wireless and optical environments. As 6G evolves, there will be a need for new, dynamic multiple access protocols that can dynamically change the type of multiple access in the wireless domain. A set of suitable optical domain configurations should be determined automatically depending on the needs of the applications and the network state. All of these protocols must be distributed and able to leverage data sets distributed across the network edge. A vision of how a new communications architecture, along with the integration of key enablers such as pervasive AI, management of communication, caching and control resources, subterahertz, and visible light communication opportunities, is presented in [11].

Advancements in 6G technology pervasively leverage intelligent communications to meet unprecedented demands and applications that are no longer beyond imagination. The entire 6G network infrastructure might well be a single ecosystem supporting real-time immersive services and high-speed access. However, this will only be realized by taking the optical transport network into account at the design phase of 6G—not as an external element but as an integral part of the entire 6G ecosystem. Only then can seamless integration among AI-driven communications, computation, and caching capabilities be fully realized.

6. CONCLUSION

With the exponentially increasing user-demand for data-intensive latency-stringent applications provided over wireless networks, the optical access network has been identified as a sustainable bridge that connect wireless data with core networks. However, optical transport networks have not evolved at the same pace as wireless networks. To this end, this paper discusses how optical access networks have been used and integrated with wireless network design and planning to aid 4G, 5G, and beyond wireless networks. This paper also discusses the drivers behind choosing optical access networks to support 4G, the challenges that 5G faces, and how optical access networks can be used as a transport solution to overcome the challenges in 5G. Finally, this paper presents key challenges in realizing 6G networks and technologies that need to be incorporated with optical transport networks to support the key requirements of 6G. Overall, this paper provides insights into key drivers and challenges in using optical transport networks to support current and emerging wireless access technologies.

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